

The (anti-)electron neutrino appearance probability is shown in Figure 1 at the DUNE baseline of 1285 km as a function of neutrino energy for several values of δ_{CP} .

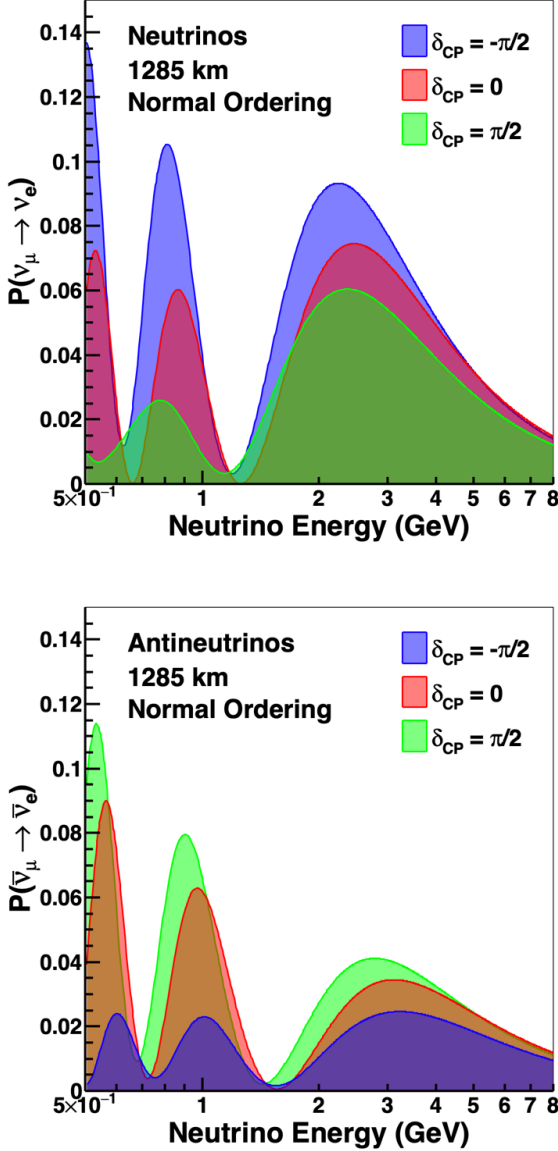


Fig. 1 The appearance probability at a baseline of 1285 km, as a function of neutrino energy, for $\delta_{\text{CP}} = -\pi/2$ (blue), 0 (red), and $\pi/2$ (green), for neutrinos (top) and antineutrinos (bottom), for normal ordering.

DUNE has a number of features that give it unique physics reach, complementary to other existing and planned experiments [19–21]. Its broad-band beam makes it sensitive to the shape of the oscillation spectrum for a range of neutrino energies. DUNE’s relatively high energy neutrino beam enhances the size of the matter effect and will allow DUNE to measure

δ_{CP} and the mass ordering simultaneously. The unique LArTPC detector technology will enhance the resolution on DUNE’s measurement of the value of δ_{CP} , and along with the increased neutrino energy, gives DUNE a different set of systematic uncertainties to other experiments, making DUNE complementary with them.

This paper describes studies that quantify DUNE’s expected sensitivity to long-baseline neutrino oscillation, using the accelerator neutrino beam. Note that atmospheric neutrino samples would provide additional sensitivity to some of the same physics, but are not included in this work. The flux simulation and associated uncertainties are described in Section 2. Section 3 describes the neutrino interaction model and systematic variations. The near and far detector simulation, reconstruction, and event selections are described in Sections 4 and 5, respectively, with a nominal set of event rate predictions given in Section 6. Detector uncertainties are described in Section 7. The methods used to extract oscillation sensitivities are described in Section 8. The primary sensitivity results are presented in Section 9. We present our conclusions in Section 10.

2 Neutrino Beam Flux and Uncertainties

The expected neutrino flux is generated using G4LBNF [5, 22], a Geant4-based [23] simulation of the LBNF neutrino beam. The simulation uses a detailed description of the LBNF optimized beam design [5], which includes a target and horns designed to maximize sensitivity to CPV given the physical constraints on the beamline design.

Neutrino fluxes for neutrino-enhanced, forward horn current (FHC), and antineutrino-enhanced, reverse horn current (RHC), configurations of LBNF are shown in Figure 2. Uncertainties on the neutrino fluxes arise primarily from uncertainties in hadrons produced off the target and uncertainties in the design parameters of the beamline, such as horn currents and horn and target positioning (commonly called “focusing uncertainties”) [5]. Given current measurements of hadron production and LBNF estimates of alignment tolerances, flux uncertainties are approximately 8% at the first oscillation maximum and 12% at the second. These uncertainties are highly correlated across energy bins and neutrino flavors. The unoscillated fluxes at the ND and FD are similar, but not identical. The relationship is well understood, and flux uncertainties mostly cancel for the ratio of fluxes between the two detectors. Uncertainties on the ratio are dominated by focusing uncertainties and are $\sim 1\%$ or smaller except at the falling edge of the focusing peak (~ 4 GeV), where they rise to 2%. The rise is due to the presence of many particles